

Article

Assessing the Potential for Intra-Day Load Redistribution in Water Intake Systems Under Different Electricity Tariff Models: A Comparative Case Study of Belarus and China

Aliaksey A. Kapanski ¹, Miaomiao Ye ^{2,*}, Shipeng Chu ^{2,*} and Nadezeya V. Hruntovich ¹

¹ Department of Power Supply, Sukhoi State Technical University of Gomel, 246746 Gomel, Belarus; kapanski@mail.ru (A.A.K.); gruntovich@tut.by (N.V.H.)

² College of Civil Engineering and Architecture, Zhejiang University, Hangzhou 310058, China

* Correspondence: yemiao008@zju.edu.cn (M.Y.); chushipeng@zju.edu.cn (S.C.)

Abstract

This article assesses the potential for intra-day redistribution of the electrical load of water intake systems under different electricity tariff models, using water supply systems in Belarus and China as case studies. It demonstrates how tariff policy influences the electrical load profile of a water intake system and quantitatively evaluates the economic effect of optimizing the operating modes of pumping equipment. The analysis is based on daily profiles of electric power and water supply. For the Belarusian water supply system, data for 2019 were considered, corresponding to the baseline operating mode without targeted load management, and data for 2023 were considered after the transition to dispatch-based control of well activation with account taken of tariff constraints (without automation tools). For the Chinese water intake system, hourly data for 2025 were used. The load redistribution potential was assessed on the basis of lagged correlation between power and water supply profiles. In addition, the *F*-index was applied as an aggregated diagnostic indicator intended for the comparative assessment of potential load transferability across technological stages, taking into account their share in total energy consumption. For the Chinese case, it was shown that the maximum correlation between water supply and electricity consumption across all technological stages is achieved near zero lag, which indicates a high adaptation of system operating modes to current demand; at the same time, the *R* values were 0.19 for reservoir intake, 0.86 for water treatment, and 0.51 for the pumping station. In the Belarusian case, for the first-lift stage, the maximum correlation is shifted by −6 h relative to zero lag, indicating a less rigid linkage of pump operation to current demand and a more inertial response of the system. A comparison of 2019 and 2023 for the Belarusian facility showed that targeted regulation of well activation and load redistribution across tariff zones reduced the total electricity cost by 1.58%, confirming the potential for further optimization of electricity consumption regimes.



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Keywords: water supply; intra-day load redistribution; water intake systems; borehole pumps; electricity tariff models; pump mode planning; load tolerance index; energy saving; economic efficiency

1. Introduction

According to international studies, electricity consumption for the operation of pumping equipment constitutes a significant share of global energy use [1]. For water utilities,

this is explained by the nearly continuous operation of pumps throughout the entire technological chain: from water abstraction and supply to maintaining pressure at points of end consumption. Consequently, urban water supply facilities are classified as large consumers of electrical energy, accounting for up to 90% of the energy balance and about 25% of total financial expenditures [2]. Against the background of increasing volatility of electricity prices, interest is growing in strategies such as demand-side management (DSM), optimization of pump operating modes, and the participation of large consumers in demand response (DR) programs [3,4]. At the same time, the industry is forming the view that energy management in water utilities includes not only optimization algorithms but also organizational and technical measures, including the development of digital infrastructure, practices for planning equipment loading, as well as methods for assessing load management flexibility under tariff and technological constraints [4]. Increasing requirements for the efficiency of water supply systems enhance the role of measures aimed at the implementation of modern energy-efficient equipment and the reduction in energy costs through optimization of pumping station operating modes and the smoothing of electrical load profiles.

In accordance with the differentiated pricing model, the final cost of electricity largely depends on the distribution of loads over the hours of the day. For this reason, the shape of the hourly power consumption profile of water intake facilities becomes a determining factor in tariff-oriented planning of pump operating schedules. However, the coincidence of water demand with periods of peak loads in the power system significantly limits the possibilities for redistributing load throughout the day and leads to the forced operation of water intakes in economically inefficient modes [5]. Over a long-term time horizon, load management reserves are formed under the influence of seasonal fluctuations, temperature conditions, and demographic characteristics, which leads to statistically significant changes in consumer water withdrawal patterns [6]. On a daily scale, water consumption is characterized by pronounced daytime and evening clusters, as well as transitional regimes in the morning hours [7]. Such a distribution of water demand significantly affects the magnitude of peak electrical power and, consequently, the total costs of the water supply organization.

The dependence of the total power demand of water intakes on the water consumption profile, along with the application of time-of-use electricity tariffs, highlights the high importance of optimizing energy consumption. In this regard, water supply systems possess a unique demand management capability due to the possibility of storing water in reservoirs and optimizing the switching of pumping loads, which makes it possible to form a desired electricity consumption profile while maintaining water supply requirements. Scientific studies show that water distribution systems can act as a demand response (DR) resource, providing the power system with controllable load due to their technological flexibility [8]. Other studies demonstrate the possibility of expanding the range of power control by equipping pumps with variable frequency drives [9]. In more recent approaches, the flexibility of water systems is considered as an object of coordinated operation with the power system, which is reflected in practice through the implementation of economically efficient pilot projects [10].

Despite the maturity of optimization problems, a more preliminary and practical question often arises in the operation of water utilities: to what extent current operations are tariff adapted, that is, to what degree the water supply profile is already aligned with the load structure, and at which stage of water transportation the potential for power shifting is formed. This issue becomes particularly relevant when comparing management models across countries and water utilities, taking into account technological and tariff-specific characteristics. In this regard, the objective of the present study is to perform a comparative analysis of the intraday relationship between water supply and electricity consumption

using the example of two water intake systems operating under different tariff conditions, as well as to propose an interpretable diagnostic metric for the preliminary assessment of potential load shiftability. The present study uses real operational data on water intake consumption and performs a comparison of normalized profiles of power demand and water supply, including an analysis of the distribution of intraday maxima relative to tariff zones and the degree of correlation across technological levels. Additionally, the F-index is introduced as an aggregated diagnostic indicator intended for the comparative assessment of potential load transferability across technological stages, taking into account their respective energy shares.

The scientific novelty of the study lies not in the formulation of a new optimization model for demand management, but in the comparison of two different water infrastructures and tariff models based on a data analysis procedure and the introduced diagnostic indicator. The practical significance of the research lies in the formation of evaluation metrics that strengthen managerial decision-making by providing visibility of the optimization potential at different technological levels. The objectives of the study include: (1) comparison of electricity pricing models and key incentives for managing electricity consumption at water supply facilities; (2) analysis of the relationship between power and water supply profiles and identification of characteristic time shifts; (3) assessment of the distribution of absolute load maxima across tariff zones; (4) quantitative evaluation of the influence of management decisions (using the example of well operation control) on changes in the daily load profile and total costs. The obtained results may be considered as a basis for the further transition to planning electricity consumption regimes in water supply systems under technological and tariff constraints.

2. Literature Review

Technological features of water supply systems make this sector a natural candidate for electricity demand management. In this regard, contemporary research highlights two complementary directions: the regulation of energy-intensive pumping equipment under the influence of tariff constraints and participation in DR programs with economic compensation for fulfilling obligations imposed by the power system. As noted in studies [3,4], the available management flexibility is determined not only by the water consumption profile but also by the technological configuration of the system itself and the set of its operational constraints. The effectiveness of applying management mechanisms is largely determined by the shape of the daily electrical load profile, which has a significant impact on the economic costs of water supply systems. The conducted review confirms that the highest costs for water utilities occur during the morning and evening hours, when electricity prices and water consumption reach their maximum values [11]. For example, according to the results of the study [12], the simultaneous operation of pumps during the evening hours leads to a sharp increase in energy costs. This effect is further amplified during the summer period due to the increase in average water demand [13,14]. At the same time, the purposeful shifting of loads from expensive tariff periods to low-tariff intervals makes it possible to achieve a noticeable economic effect [15,16]. Similar results are presented in [17], where the authors achieved a reduction in total operating costs by shifting part of the demand from expensive hours to cheaper ones and by reducing overall electricity consumption through energy-saving measures.

In more formalized approaches, the participation of water supply systems in demand management is described as an optimization problem that considers the combined flexibility of pumping equipment, reservoirs, and the hydraulic network. In [18], a model is proposed for planning the participation of water distribution systems simultaneously in demand response programs and power regulation services, taking into account the

constraints of the pipeline network. In [19], the concept of the energy flexibility of a water distribution system is introduced in the context of day-ahead power system operation, which significantly expands the understanding of DSM from simple load shifting within tariff zones to the planning of load management. In practical applications, the development of such studies has led to the emergence of forecasting and optimization models for water intake management [20,21], based on scheduling pump operation while considering forecasted water demand and the current reservoir filling level [22,23].

Alongside the development of methods for optimizing the operating modes of pumping stations, a separate research direction has emerged in the scientific literature that focuses on assessing the potential for demand management. For example, study [24] proposes a model for evaluating controllable power and determining an optimal bidding strategy to maximize the revenue of water utilities. The practical effectiveness of such approaches is demonstrated by the results presented in [25], which show that shifting 3.8 MW of load to off-peak periods of the power system for a large pumping system ensured the payback of automation investments in less than six months. This category of studies emphasizes that an appropriate flexibility metric requires explicit consideration of technological constraints and the objective function of optimization. In problems involving participation in electricity markets, studies addressing risk management under conditions of uncertainty in water demand are of particular importance. For example, Mkireb et al. [26] examines the participation of drinking water supply systems in Demand Response programs under uncertainty in water consumption, using probabilistic constraints and evaluating economic benefits together with the risk of non-compliance with contractual obligations.

Price-dependent demand management is identified as a separate research direction in which load management problems are considered while accounting for the complex constraints of water and energy systems. In [27], water supply pumping stations are considered as consumers with flexible electrical loads within a planning problem under uncertainty in both water demand and electricity prices. The problem is formulated as a probabilistic optimization model that enables the formation of water pumping schedules and real-time control strategies without violating the constraints of water and electrical networks. Developing this approach, Stuhlmacher et al. [28] investigates the possibility of using water supply pumping stations to simultaneously provide frequency regulation and voltage support services to the power system while optimally controlling water pumping. At the same time, price-based demand management mechanisms are also considered an effective tool for improving energy efficiency. In [29], it is shown that optimizing pump operation according to wholesale electricity prices reduces water supply costs and can increase the integration of wind generation by shifting load to periods of low electricity prices. These results directly link the economic benefits of pump control with the tariff structure of the power system and highlight the need to implement intelligent control tools for managing the operating modes of water supply systems.

In this regard, advanced water supply control centers integrate analytical modules, including those based on artificial intelligence methods, into a unified dispatching platform to support decision-making [30]. One approach to implementing such systems is presented in [31], where a real-time pump control method based on deep reinforcement learning is proposed, enabling the optimization of operating modes using instantaneous hydraulic data from the system. Further developing this approach, Ma et al. [32] proposes a physics-informed multi-agent deep learning algorithm for optimizing the operating schedule of pumping stations while accounting for the constraints of the water distribution network. A practical demonstration of the use of water distribution network flexibility is presented in [10], where optimization of operating modes was implemented at a real facility in Belgium, taking into account demand forecasts and electricity prices. Thus,

contemporary research demonstrates a transition from individual optimization algorithms to comprehensive intelligent control systems for water supply.

As the literature review shows, forecasting and optimization methods for water supply management are actively developing worldwide and receive considerable attention in the scientific community. At the same time, the resulting algorithms differ across countries and largely depend on national regulatory frameworks and incentives aimed at smoothing load profiles. The available literature lacks studies devoted to a comparative analysis of water supply systems with different levels of adaptation of electricity consumption regimes to water demand profiles. In contrast to the studies discussed above, the present research focuses on the preliminary diagnosis of the consistency between water supply and electricity consumption in two systems with different infrastructure and tariff conditions. The results can be used as a screening basis for assessing the potential for demand management prior to further optimization and formalization using approaches similar to those presented in [24], as well as for evaluating the readiness of utilities to participate in more advanced DSM/DR mechanisms described in the reviews [3,4].

3. Methodology

The study is based on hourly data on the operating modes of Belarusian and Chinese water supply systems. The overall design of the study includes: (1) an examination of the infrastructural features of water intakes; (2) data collection and standardization; (3) an analysis of tariff similarities and differences between the studied countries; (4) a comparison of water supply and electricity consumption profiles; (5) a determination of the possibilities for regulating the load schedule and the prerequisites for scheduling pumping equipment operating modes. To objectively assess the load manageability, operational and tariff constraints of the water supply system (charges for capacity and energy, time intervals of the power system) were taken into account.

3.1. General Characteristics of the Research Objects

3.1.1. Features of the Belarusian Water Supply Infrastructure

In Belarus, the study was conducted at a regional water supply company, which comprises 21 divisions in the Gomel region, producing, purifying, and transporting water for urban and rural consumers. Figure 1 shows the company's territorial structure, displaying its branches and their water production levels.

The map in Figure 1 highlights the administrative center, the city of Gomel, with a population of over 600,000. The city's water supply system is based on five interconnected water intake sources of varying capacity. The primary focus of the study is the Sozh water intake, part of the Gomel city network and one of the city's largest water supply sources, with an annual production volume of over 18 million m³. The water intake includes 40 artesian wells with a total installed capacity of over 960 kW and a total hourly flow rate of 3750 m³/h. The structure of the annual balance of electricity consumption is characterized by the predominance of the costs of the first water lift, which accounts for up to 69% of the total annual electricity consumption of the water intake.

The second-lift pumping station, part of the water intake facilities, is equipped with six units with a total installed capacity of 2115 kW. The station's operating mode is based on a staggered pump activation based on the city's water demand: one unit operates at night, while two pumps are activated during peak water consumption. Some of the pumping equipment is used as a reserve, which results in a low electrical load at the station compared to design parameters. The second-lift station's energy consumption accounts for approximately 31% of the annual volume, including general production needs ($\approx 2\%$) and process costs for filter cleaning ($\approx 8\%$). Figure 2 presents the distribution of energy

indicators of the water intake by technological operations and the structure of electricity consumption. The bar chart shows the structure of the total installed power capacity by technological stages based on the nominal equipment characteristics, while the pie chart reflects the actual distribution of electricity consumption between the first and second water lift stages. Due to the existing limitations of the electricity metering system, the electricity consumption associated with water treatment processes is included within the second lift stage.

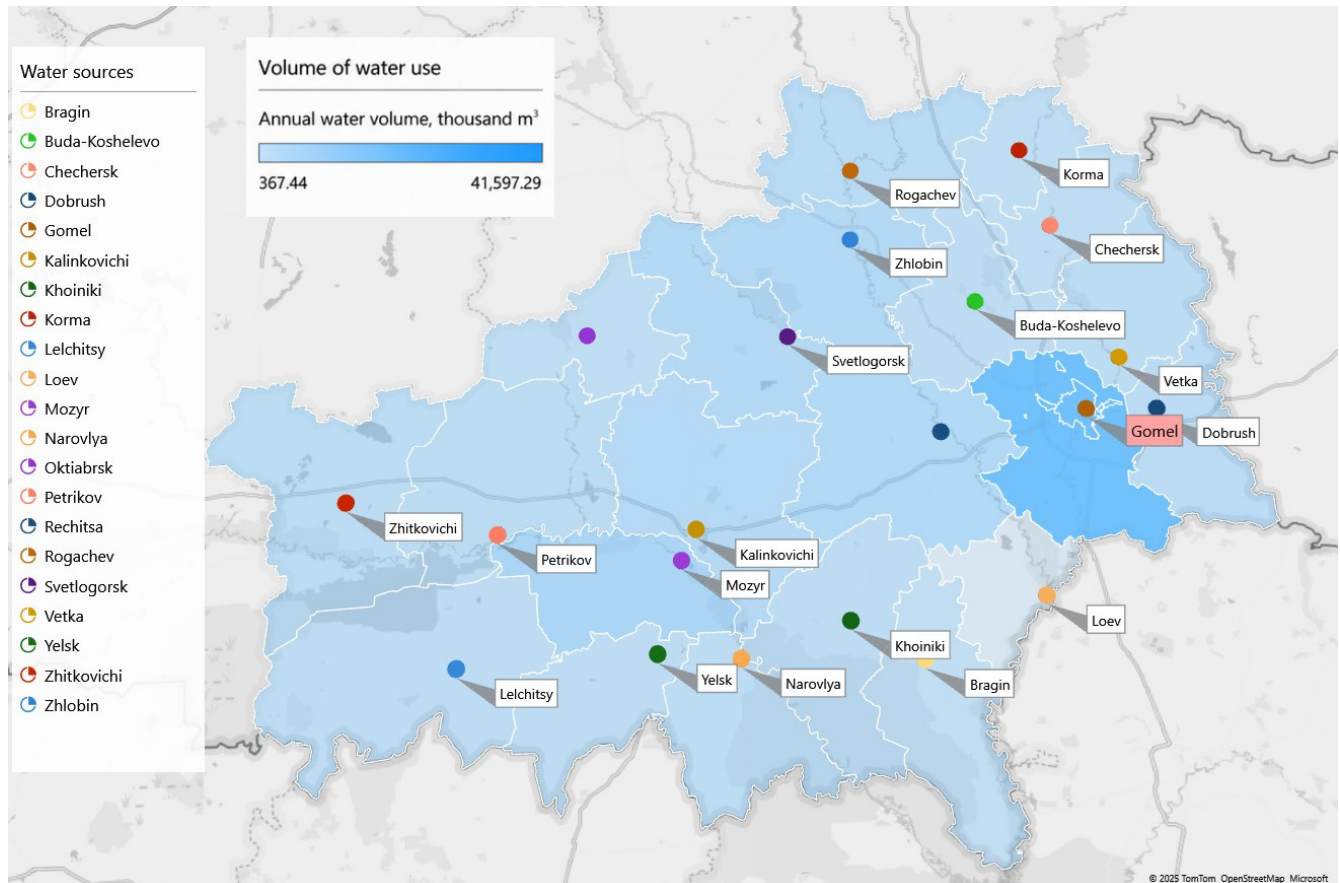


Figure 1. Territorial distribution of water supply companies in the Gomel region of Belarus and their capacity.

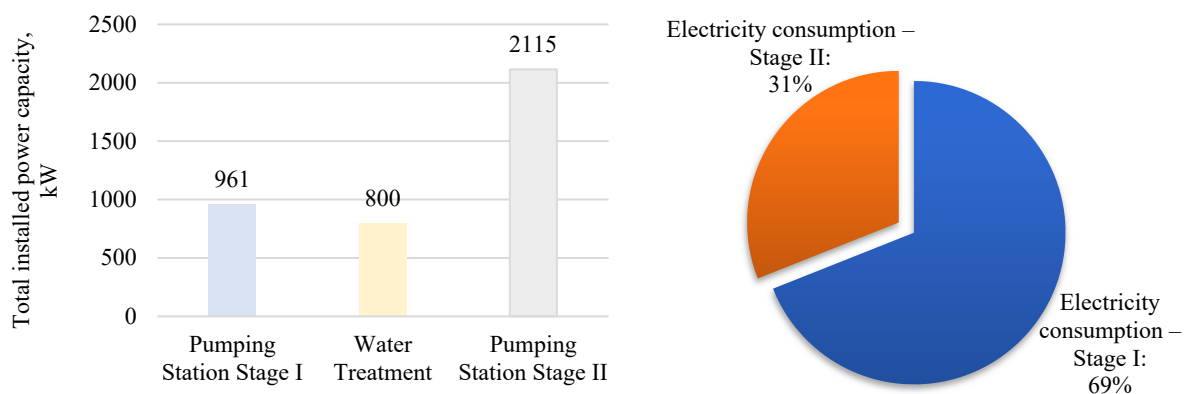


Figure 2. Distribution of installed capacity and actual balance of electricity consumption of the Belarusian water intake by technological operations.

Figure 2 shows that the first lift level accounts for the largest share in the electricity consumption structure of the water intake system. Owing to the presence of storage reservoirs with a total capacity of 30,000 m³, this stage has the greatest potential for intra-day load shifting while maintaining the required water balance. At the treatment stage, load shifting is more limited but can be achieved by scheduling filter backwash operations and operating backwash pumps during off-peak periods in compliance with technological constraints. Given the total installed capacity of 800 Kw, shifting backwash operations to nighttime hours can significantly reduce peak loads during periods of maximum stress on the power system. The second lift level is more strongly tied to real-time consumer demand and distribution network pressure requirements; therefore, its contribution to load shifting is limited and can be achieved mainly through adaptive pressure control, rather than by fully rescheduling pumping operations in time.

The principal technological scheme shown in Figure 3 illustrates the sequential movement of water and the distribution of energy flows within the water intake facilities, from groundwater abstraction by wells to delivery into the urban distribution network. For the first lift, the possibility of load schedule shifting is constrained by the permissible reservoir levels (operating range 2.3–4.84 m), as well as by the requirement to maintain a minimum fire-protection water reserve and to comply with the upper filling limit, which together define the allowable boundaries for changing pump operating regimes. At the treatment stage, flexibility is further regulated by the mandatory backwashing of 14 open filters (6 × 12 m) performed according to the technological schedule. On average, for the system under consideration this corresponds to approximately 486 backwash cycles per year and about 200 thousand m³ of backwash water annually. In addition, the potential for operational rescheduling is limited by the duration of the filtration cycle, the quality of the raw water, and the requirement to keep a sufficient number of filters in operation to ensure reliable hydraulic throughput of the treatment facilities.

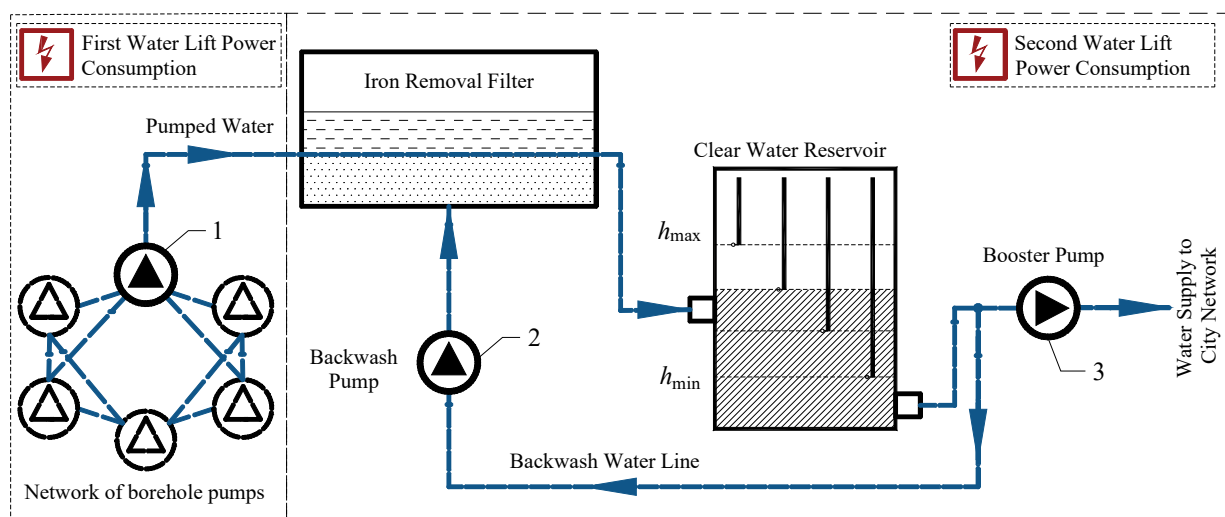


Figure 3. A simplified representation of the Belarusian water intake infrastructure. (1—borehole pump; 2—backwash pump; 3—second water lift pump.)

3.1.2. Features of China's Water Supply Infrastructure

Figure 4 presents a schematic diagram of the water supply system serving a city in East China with a population of over 2 million and a daily water supply of 400,000 cubic meters (m³/d). The figure provides a simplified representation of the water intake infrastructure, where elements are grouped according to electricity metering boundaries, including

pumping equipment and other process electricity consumers within the corresponding technological stage.

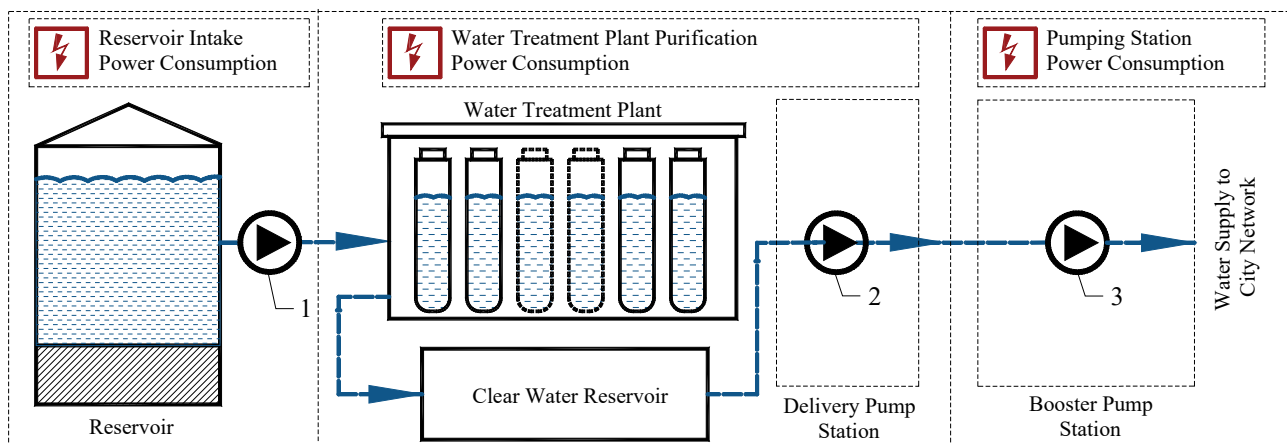


Figure 4. A simplified representation of China's water intake infrastructure. (1—Supply Pumping Station; 2—Delivery Pump Station; 3—Booster Pump Station.)

This Chinese water supply system features a linear water transportation structure with a surface water intake source from a reservoir: raw water is conveyed to the water treatment plant through a 6.7-km DN2000 main water pipeline, and the water treatment process includes preliminary chlorination, coagulation, clarification and filtration. After purification, the water is pressurized and pumped out from the pressure pumping station, then transported via a 10.5-km DN1600 distribution pipeline to the booster pumping station, which distributes water to each water consumption node of the city while maintaining a preset network pressure. In addition, continuous metering of electricity consumption and water supply volume is implemented at each technological process stage of the system, with the collected data serving as the research data source for this study. The figure gives a simplified representation of the infrastructure of this Chinese water intake system, and its functional components are grouped and classified by energy metering units for statistical analysis.

Figure 5 shows the distribution of maximum power and electricity consumption of water intake by technological operations.

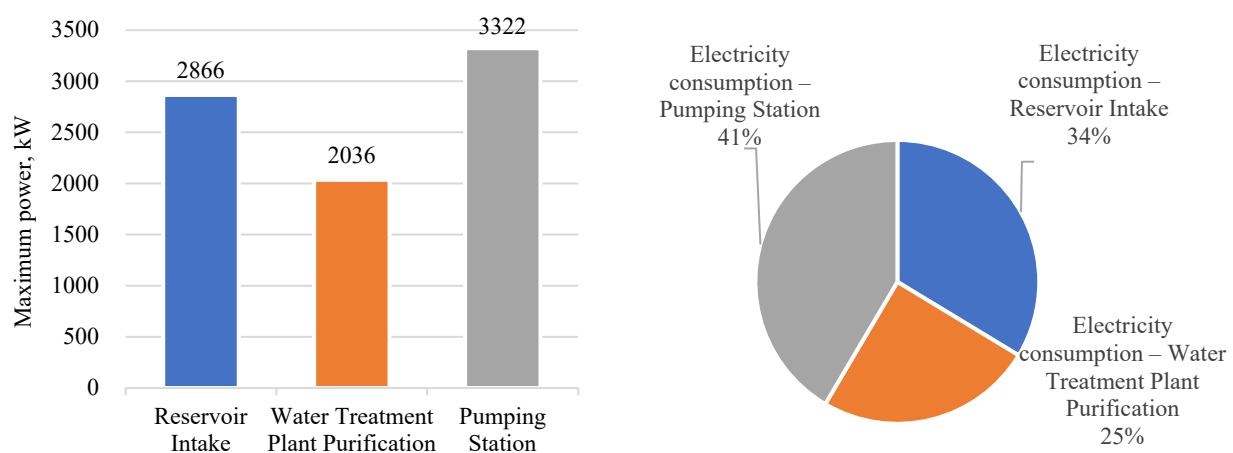


Figure 5. Distribution of maximum hourly power and electricity consumption of Chinese water intake by technological operations.

A preliminary analysis of the energy structure revealed that booster pumping stations account for over 41% of the system's total costs. Energy consumption for water extraction from reservoirs accounts for 34%, while the water treatment plant accounts for only a quarter of the system's total energy consumption. The total hourly peak capacity for all identified water supply groups is approximately 8.1 MW.

3.1.3. Comparability Analysis of Belarusian and Chinese Water Supply Systems

Although the Sozh water intake system in Gomel, Belarus, and the water supply system of a major city in East China exhibit inherent differences in urban characteristics, water source types, and infrastructure configurations, they are highly consistent in key aspects such as system attributes, electricity tariff mechanisms, and research dimensions—laying a solid foundation for cross-country comparative analysis.

1. Consistency in System Attributes and Processes

Both systems serve the production and domestic water needs of core urban areas, featuring the typical urban water supply process flow of water intake, water treatment, and pressurized transmission/distribution. As major regional electricity consumers, they qualify as large-scale water supply facilities: the Sozh water intake system in Belarus has a total installed capacity of approximately 3800 kW, while the Chinese water supply system reaches an hourly peak capacity of 8000 kW. The core technical logic underlying pump station operation and load regulation is universally applicable to both.

2. Homology of Electricity Tariff Mechanisms

Both water supply systems adopt a dual-tariff model (capacity charge + energy charge) combined with time-of-use (TOU) pricing (peak, off-peak, and valley periods). Both guide the temporal shift of electrical loads through tariff time zone division, with capacity charges serving as a critical component of total electricity costs and a core indicator for cost control. This similarity in tariff mechanisms provides a unified basis for comparing the impact of electricity pricing on energy management strategies.

3. Alignment in Management Needs

Both systems prioritize core energy management goals, including optimizing pump station operation schedules, reducing electricity costs, and achieving inter-temporal load transfer. They also face the common challenge of low economic efficiency caused by overlapping water demand and power grid peak loads. The research findings are mutually referable, offering practical insights for energy optimization in water supply systems across different countries.

The selected water supply systems in Belarus and China demonstrate high direct comparability in terms of large-scale public water supply system attributes, TOU dual-tariff mechanisms, and core energy management objectives—making them suitable subjects for cross-country comparative analysis of energy management patterns in water supply systems.

3.2. Sources of Statistical Information and Data Processing Methods

3.2.1. The Belarusian Water Supply System

At the water intake site under study, the power supply system is also structured by process stages. At the first lift level, energy consumption is metered for the entire group of 40 artesian wells. The second lift electricity metering circuit includes both booster pumps and general-purpose process electricity consumers located within the Water Treatment Station (WTS). The automated metering system provides highly accurate measurement in half-hour increments. The data used has commercial status, which is used in settlements with the energy supply organization.

For the description of water supply operating regimes, the study used data on water delivery to the urban distribution network (Figure 3). A specific feature of the considered water intake facility is the absence of a digital system for recording water demand. The supplied water volumes are recorded hourly by the operator based on flow meter readings and entered into a dedicated machine room operational log. The correctness of the records is controlled through regular verification of daily and monthly water balance calculations of the system, which makes it possible to identify and correct potential discrepancies in accounting. Despite the partially manual nature of the registration process, the information undergoes operational verification and is used by the utility for routine operational control, which allows it to be applied in the analysis of aggregated relationships between water supply and energy consumption. The research methodology included the digitization of the original records, their transfer into electronic format, and subsequent integration into an information database for further analysis. Microsoft Excel spreadsheet models (v2021) were used for data processing, structuring, and initial data presentation. Before analytical processing, the statistical data were then checked for outliers and gaps in the Python software environment (v3.11) using the Pandas and NumPy libraries. A brief description of the data is provided in Table 1.

Table 1. Characteristics of the studied statistical data of the Belarusian water source.

Data Parameter	Data Source	Measurement Step	Data Volume	Research Period	Note
Electricity for the first water lift	Commercial electricity metering system	30 min	35,040	2019, 2023	Summary data on the operation of borehole pumps
Electricity for the second water lift	Commercial electricity metering system	30 min	35,040	2019, 2023	Includes second water lift pumps and WTS process needs
Water supply to the city network	Pumping station operator's log	1 h (digitized)	17,520	2019, 2023	The values were recorded manually. The data have been converted to digital format.

3.2.2. The Chinese Water Supply System

The data source for the Chinese water source was hourly statistics obtained from technical metering systems for the period from 1 January 2025 to 31 December 2025. Electricity, water supply, and pressure data were recorded at a one-hour interval and presented by key process operations. Specifically, the study collected, aggregated, and presented in a summary table a historical data set for the reservoir, water treatment, and pumping stations for transmission and pressure boosting. Water supply to the urban distribution network was also recorded at an hourly interval. Table 2 provides a general description of the analyzed statistical data.

Table 2. Characteristics of the studied statistical data of the Chinese water source.

Data Parameter	Data Source	Measurement Step	Data Volume	Research Period	Note
Electricity by technological stages	Electricity metering system	1 h	8760	from 1 January 2025 to 31 December 2025	The data was recorded at the following stages: water intake from the reservoir; water treatment; operation of pumping stations for transportation and pressure increase.
Water supply to the city network	Electricity metering system	1 h	8760	from 1 January 2025 to 31 December 2025	

3.3. Data Reconciliation and Comparability of Research Objects

The Belarusian and Chinese facilities are treated as two contrasting practical case studies and are not compared in terms of population size, the share of industrial water consumption, or the level of urbanization. The comparison is carried out at the level of daily profile shapes, the coupling between water delivery and power consumption («water supply–power»), and the distribution of load maxima with respect to tariff time zones. To ensure a correct interpretation, the analysis relies on normalized indicators of water intake operating regimes. To implement this comparison and ensure the correctness of profile-based metrics, the original time series were harmonized to a common time step. The need for this procedure is due to differences in sampling resolution and the heterogeneity of measurements across technological stages. For the Belarusian water intake, the original half-hourly measurements were aggregated into an hourly time interval to ensure consistency with the temporal resolution of the Chinese dataset. Table 3 presents the characteristics of the data set and the normalized indicators used in the analysis of water supply systems.

Table 3. Dataset characteristics and normalized metrics used in water supply system analysis.

Indicator	The Belarus (2023)	China (2025)
Data period (start–end)	from 1 January 2025 to 31 December 2025	from 1 January 2025 to 31 December 2025
Measurement interval	30 min	1 h
Number of days	365	365
Average daily water supply (m ³ /day)	51,390	317,571
Average daily electricity consumption (kWh/day)	27,030	157,102
Specific energy consumption at all levels (kWh/m ³)	0.526	0.495
Mean interval-averaged power (kW)	1126	6546
Peak interval-averaged power (kW)	1632	10,427
Peak-to-average power ratio	1.449	1.593
Mean outlet pressure (kPa)	415	527

As an initial preprocessing step, all data were first standardized to a half-hourly resolution to maintain internal consistency within the Belarusian dataset. However, to ensure statistical accuracy of the analysis and eliminate the endogeneity of Belarusian and Chinese data, an hourly time interval of $\Delta t = 1$ was adopted as the basic time resolution for the subsequent cross-country comparative analysis.

Thus, electricity consumption data recorded in 30-min increments were aggregated to an hourly value by summing the energy:

$$E_h = E_{h,0} + E_{h,1}, \quad (1)$$

where $E_{h,0}$, $E_{h,1}$ —half-hour intervals of energy supply for index $h,0$ within 0–30 min and for index $h,1$ within 30–60 min.

In the comparative analysis, total consumption included dividing the electricity into i -th process stages $E^{(i)}$:

$$E_{tot} = \sum_i E^{(i)}. \quad (2)$$

In this case, the average power was determined as a function of energy consumption in a time interval Δt :

$$P_{tot} = \frac{E_{tot}}{\Delta t}. \quad (3)$$

Data accuracy was verified by checking the values during the reverse aggregation of data from half-hourly to hourly values, as well as by the total sum of the integral indicator values. It is worth noting that for the Belarusian water intake facility, the initial data was compiled from operational logs, so the dataset was pre-digitized and structured as a time series for two full years (2019 and 2023), corresponding to the regimes before and after the implementation of targeted well activation regulation. For the Chinese site, data was obtained from a digital monitoring system, resulting in a consistent set of hourly water supply and electricity consumption data for 2025 available for analysis.

3.4. Tariff Models for Electricity Cost Calculation

3.4.1. Electricity Payment Features in Belarus

In the Republic of Belarus, the mechanism for calculating the cost of electricity for water supply consumers depends on the connected load capacity, which is estimated based on the total transformer capacity, and the availability of an automated commercial electricity metering system [33]. For transformer capacities less than 750 kVA, a single-rate payment system or simple tariff differentiation by time zone (right-hand side of Figure 6) is used on an alternative basis. In this case, the final cost of electricity is determined by multiplying the consumption volume by the current tariff, while the actual power consumption is not considered as a separate economic component:

$$C_{srt} = c \cdot E_{tot}, \quad C_{tou} = \sum_i c_i \cdot E_i \quad (4)$$

where C_{srt} , C_{tou} —accordingly, pricing is based on a single-rate tariff and time-of-use pricing; c —electricity tariff with single-rate payment; c_i —electricity tariff in the i -th time zone; E_i —electricity in the i -th time zone.

For enterprises with installed capacity over 750 kVA, a two-part tariff model is used, in which payment consists of two components: an electricity charge and a charge for half-hourly power. The charge for power is calculated based on the highest half-hourly load value recorded during the established peak hours of the power system (morning peak 8:00–11:00, evening peak 19:00–23:00). This economic model encourages enterprises to adjust equipment operating schedules to reduce power peaks when calculating the cost of electricity:

$$C_{atpt} = a \cdot P_{\max} + b \cdot E_{tot} \quad (5)$$

where C_{atpt} —total electricity cost under the two-part tariff model for the considered billing period; a , b —tariff rate for capacity and electricity respectively; $P_{\max}$ —the actual value of the highest power for the billing period, falling during the hours of maximum load of the power system; E_{tot} —volume of electricity consumed.

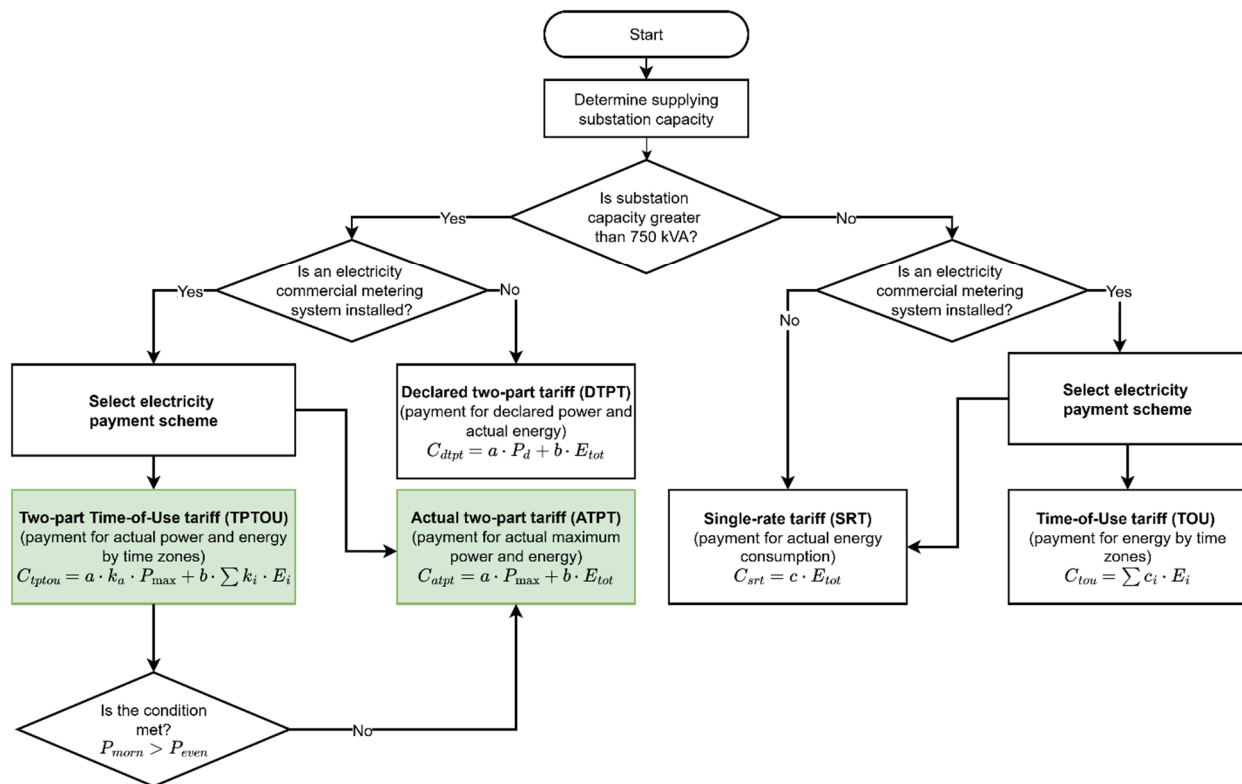


Figure 6. Tariff system for payment for electricity in the water supply system of the Republic of Belarus.

In addition to the basic two-rate model, enterprises with a connected capacity of over 750 kVA can switch to time-of-use pricing. In the Republic of Belarus, fixed time boundaries for tariff zones are applied: a peak zone (from 8:00 to 11:00), coinciding with the period of the morning maximum load on the power grid; a semi-peak zone (from 6:00 to 8:00, from 11:00 to 23:00); a night zone (from 23:00 to 6:00), corresponding to the interval of minimum load on the power grid. This model creates conditions for more flexible load management during the day and economically incentivizes consumption reduction during hours of high energy costs:

$$C_{tou} = a \cdot k_a \cdot P_{max} + b \cdot (k_{off} \cdot E_{off} + k_{sh} \cdot E_{sh} + k_{pk} \cdot E_{pk}) = a \cdot k_a \cdot P_{max} + \sum_i k_i \cdot E_i, \quad (6)$$

where k_a —a reduction coefficient to the basic rate of the two-part tariff, set at 0.5; k_{off} , k_{sh} , k_{pk} —dimensionless tariff adjustment coefficients for the night, semi-peak and peak zones, respectively; E_{off} , E_{sh} , E_{pk} —electricity in night, peak and semi-peak zones; k_i —dimensionless tariff coefficient for the i -th time zone.

A key feature of this model is that to retain eligibility for differentiated payment, the evening peak P_{even} (6:00 PM to 9:00 PM) must not exceed the morning peak P_{morn} (8:00 AM to 11:00 AM). Otherwise, the consumer automatically switches to the standard two-rate system (Formula (5)), which increases the share of payment for capacity. It should be noted that the differentiated tariff allows for a halving of the cost of capacity compared to the basic model, but is accompanied by a significant increase in the cost of electricity during peak demand hours. In the study conducted for a Belarusian water supply organization, one of the important indicators is the average load, the excess of the evening peak over the morning peak:

$$\Delta_d = \max_{18:00-21:00} P(d) - \max_{08:00-11:00} P(d), \quad (7)$$

where $P(d)$ —half-hour active power in the time interval of the power system maximum; d —day under study.

Figure 6 presents a generalized view of the tariff system for payment for electricity in the Republic of Belarus, with blocks that reflect possible options for its application for the water supply system in question, based on a transformer capacity of more than 750 kVA.

3.4.2. Peculiarities of Payment for Electricity in China

In China, electricity prices for industry are determined based on the type of industry, electricity consumption regime, regional distribution, consumption periods, and the chosen pricing methods [34,35]. The multi-tiered structure of the Chinese model is generally comparable to the Belarusian one in the basic logic of separating payments for capacity and electricity, but it has a more detailed breakdown into economic components. As a result, the final cost of electricity is determined by the sum of the following components [36]: the fee for electricity consumed from the grid $C_{e,tot}$, the transmission and distribution fee $C_{t,d}$, the loss and system operation fee $C_{l,s}$, and the state fund surcharge C_f . The generalized formula for electricity payment is:

$$C = C_{e,tot} + C_{t,d} + C_{l,s} + C_f \quad (8)$$

In the cost structure, payment for consumed energy $C_{e,tot}$ accounts for approximately 60% of the final cost. Transmission and distribution costs average 30% of the total electricity price and are generated from the authorized revenue of grid companies. Payments for transmission line losses and power system operation, as well as state fund surcharges, account for approximately 5%, respectively. It is worth noting that in the Belarusian electricity payment model, all additional components related to the operation of the grid infrastructure and line losses are included directly in electricity tariffs and are not billed separately.

In China, similar to the Belarusian model, the electricity payment system depends on the installed capacity. The total capacity of transformers is also used as a criterion. A fixed rate is applied based on actual electricity consumption, regardless of the equipment capacity, with a single tariff applied to consumers with transformers of 100 kVA and below. In the capacity range from 100 to 315 kVA, consumers can choose between a single-rate or two-rate payment model. For consumers above 315 kVA, which includes the water supply system under study, a two-rate model is applied, including a charge for both capacity and electricity:

$$C_{e,tot} = C_p + C_e \quad (9)$$

where C_e —electricity bill; C_p —power charge determined by the formula:

$$C_p = c_p \cdot P_{\max} \quad (10)$$

where c_p —tariff for electric power; $P_{\max}$ —maximum consumed active power.

A distinctive feature of the Chinese tariff model is the ability to choose a capacity payment mechanism, which can be calculated in two ways: based on transformer capacity or on maximum power consumption. In the latter case, the consumer can pay based on the contractual value or the actual recorded maximum consumption. It is crucial to note that the maximum capacity is determined over the entire billing period and is not limited to fixed hours in the power grid, as is typical of the Belarusian electricity payment model.

In the two-component model, electricity charges are calculated with differentiation by time zones determined by regional characteristics. In most regions, as in the Belarusian model, a three-zone tariff structure is used, which includes peak, semi-peak, and off-peak periods. The typical interval boundaries correspond to the morning and evening peak loads of the power system (from 8:00 to 11:00 and from 18:00 to 21:00), while the off-peak

period covers the time period from 23:00 to 6:00. The rest of the day is a semi-peak period. It should be noted that these time intervals are generally comparable to the differentiated zones used in the Belarusian model. The exception is the period of the evening peak load of the power system, which in the Republic of Belarus is regulated through a capacity fee and is not related to the peak tariff interval. For the Chinese case study considered in this paper, the water supply system operates under a 10 kV two-component electricity tariff with time-of-use pricing. The peak tariff periods (08:00–11:00 and 18:00–21:00) correspond to an electricity price of ¥0.992/kWh. The off-peak period (23:00–06:00) has a tariff of ¥0.294/kWh, while the remaining hours correspond to a semi-peak tariff of ¥0.643/kWh. Comparative indicators of the time tariff zones of the Chinese and Belarusian electricity payment models are shown in Table 4.

Table 4. Comparative indicators of time tariff zones of the Chinese and Belarusian models of payment for electricity.

Name of the Comparison Indicator	China's Electricity Payment Model	The Belarusian Electricity Payment Model	Note
Night zone	Usually 23:00–06:00 (7 h)	23:00–06:00 (7 h)	The time boundaries of the night zone are completely identical
Peak Zone	Usually 08:00–11:00 and 18:00–21:00 (6 h)	08:00–11:00 (3 h)	The morning peak is the same in both models; the Chinese system additionally highlights the evening peak as the peak energy zone.
Off-Peak Zone	Other hours (11 h)	06:00–08:00 and 11:00–23:00 (14 h)	In the Belarusian model, the evening period belongs to the off-peak (semi-peak) zone, while in the Chinese system, part of these hours is included in the peak zone.
Power Payment Zone	Absolute actual maximum active power recorded for the billing month	Maximum active power recorded during the set hours of maximum power system demand (08:00–11:00 and 19:00–23:00)	In the Chinese model, the calculated capacity is not time-bound, whereas in the Belarusian system it is strictly linked to specific hourly intervals.

Comparison of the time-of-use zones in Table 4 allows a direct interpretation of the economic feasibility of shifting pumping operating modes to off-peak hours. In both models, the night zone coincides (23:00–06:00); however, in the Chinese system an additional evening peak energy tariff period is defined, which strengthens the incentive to shift energy-intensive operations (including backwashing and part of water lifting in the presence of storage reservoirs) from the 18:00–21:00 interval to nighttime hours. In the Belarusian model, the energy peak window is shorter (08:00–11:00), and the economic incentive to limit the load during evening hours is determined by the power charge. The assessment of the shifting effect is based on the tariff difference between peak and night periods, taking into account the volume of shifted energy, as well as on the reduction in the power charge through decreasing the maximum active power, which will be discussed in Section 4.3.

3.5. Methods of Comparative Analysis

The comparative analysis was performed using two complementary scenarios: a within-country comparison for Belarus for 2019 and 2023 to identify changes in the electricity consumption profile due to the pump management strategy; and a cross-country comparison between Belarus (2023) and China (2025) to assess differences in the shapes of daily water and energy consumption profiles and the potential for load shifting by time of day. To ensure a fair comparison of the results for the Belarusian water supply system, the data were normalized to a single scale. The initial assessment of the relationship between water supply and electricity consumption was carried out using Pearson correlation analysis:

$$r_{Q,P} = \frac{\sum_{i=1}^n (Q_i - \bar{Q})(P_i - \bar{P})}{\sqrt{\sum_{i=1}^n (Q_i - \bar{Q})^2} \sqrt{\sum_{i=1}^n (P_i - \bar{P})^2}}, \quad (11)$$

where P_i —is the average power during the i -th interval Δt ; Q_i —the water supply volume normalized to the interval Δt ; n —number of intervals.

The comparison of consumption profiles was performed using typical daily dynamics with a consistent discrete time step Δt . The construction of daily power profiles was based on averaging all values within the calculated intervals Δt :

$$\bar{P}_k = \frac{1}{m_k} \sum_{j=1}^{m_k} P_j \quad (12)$$

where $\bar{P}_k$ —average power in the k -th hour interval; P_j —actual power values; m_k —number of points within an interval.

When analyzing the intraday distribution of maximum loads, two different indicators were considered: the absolute maximum and the calculated maximum. The absolute maximum was used to identify the hours of day that had the highest active power values. To generate a frequency profile of absolute maximums for each day d , a time series was considered $P_d(t)$, after which the highest value was determined and the time of its occurrence was recorded $t_{d,max}$. The maximum load was defined as:

$$P_{d,max} = \max_{t \in d} P_d(t), \quad (13)$$

The calculated profile of maxima was determined by averaging all maxima in the k -th interval and was used to estimate the stable concentration of loads by day zones:

$$\bar{P}_{d,max} = \max_k \bar{P}_d(t). \quad (14)$$

Next, the frequency of peak occurrence in hour intervals were calculated. The frequency of occurrence was determined by the number of days on which the peak occurred within the time interval t_j :

$$f_j = \sum_{d=1}^D (t_{d,max} = t_j). \quad (15)$$

A comparative analysis of daily water and electricity consumption profiles was performed by normalizing to the mean value:

$$P^*(t) = \frac{P(t)}{\bar{P}}, \quad Q^*(t) = \frac{Q(t)}{\bar{Q}}, \quad (16)$$

where $\bar{Q}$, $\bar{P}$ —average value of water consumption and electric power for the period under review.

3.6. Diagnostic Analysis of Load Transferability

In water supply systems, energy consumption is generally determined by the volume of water supplied. At the same time, the structure of energy costs includes general-purpose operations for which the direct dependence on the current flow rate is weak or absent. For this reason, a weak relationship between water supply and power consumption may indicate either potential load transferability or process inertia associated with water storage. Figure 7 shows that the change in reservoir level is almost linearly related to the difference between water abstraction from wells and water supply to consumers, which leads to a temporal decoupling between pump operation and actual water demand.

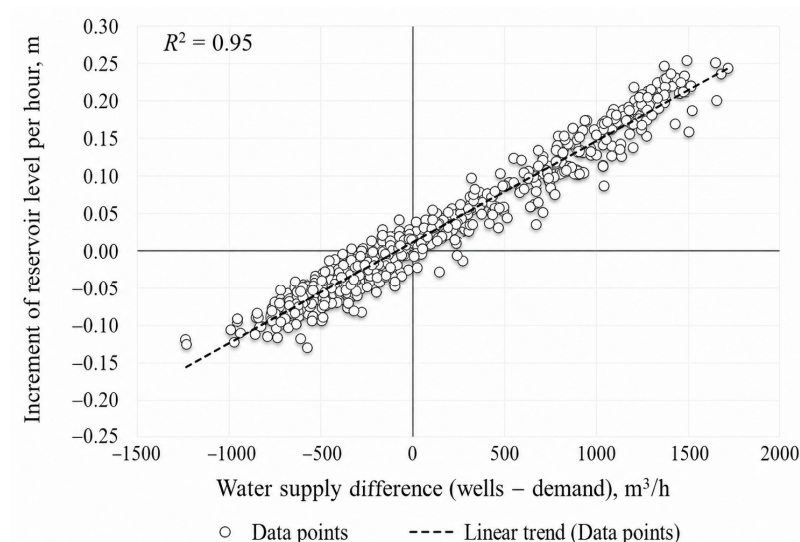


Figure 7. Relationship between the water balance and changes in reservoir water level.

Under these conditions, correlation analysis is used to assess the degree of consistency between water supply and energy consumption at different technological levels. For the initial diagnosis, the *F*-index, lag correlation analysis, and assessment of tariff-driven shifts of energy relative to water supply are applied. This combined approach is aimed at identifying the initial potential for load management based on the analysis of the residual consistency between the energy profile and water demand. At the same time, the actual optimization and confirmation of the feasibility of the identified potential require a detailed study of the technological process using classical analytical methods and considering operational constraints [8,10,24].

3.6.1. F-Index for Assessing Potential Load Transferability

The proposed diagnostic indicator, the *F*-index, reflects the share of energy that does not require strict synchronization with water supply. The quantitative indicator is calculated based on the analysis of the correlation between energy consumption and water demand at different technological levels, taking into account the contribution of each level to the total energy consumption:

$$s_i = \frac{\sum_t E_i(t)}{\sum_t E_{tot}(t)}, \quad (17)$$

where s_i —energy share of the i -th technological level of the water supply system, $s_i \in [0, 1]$; E_i —electricity consumption of the i -th technological level; E_{tot} —total electricity consumption.

As a result, an integrated estimate of the share of energy consumption at technological levels with weak coupling to water supply is obtained. The corresponding indicator is defined as follows:

$$F = \sum_{i=1}^m s_i(1 - |r_i|) = \sum_{i=1}^m F_i, \quad (18)$$

where r_i —correlation coefficient at the i -th technological stage; m —number of technological levels.

In accordance with the infrastructure characteristics (Section 3.1.1), two technological levels ($m = 2$) were considered for the Belarusian water intake, including the energy consumption for water abstraction (first lifting stage) and water supply to the distribution network (second lifting stage). This is determined by the technological features of the system and the structure of energy accounting: water treatment in the considered Belarusian system accounts for a minor share of the total energy balance. In terms of backwashing, a simplified aeration method is used, and energy consumption is mainly associated with the operation of backwash pumps, which occur with low frequency. For the Chinese infrastructure of the studied facility (Section 3.1.2), three technological levels ($m = 3$) were identified.

3.6.2. Lag Analysis of the Delay Effect

To avoid misinterpreting load transferability that may actually be caused by delayed tracking of water supply, the lagged correlation is calculated with a time shift step τ :

$$r_{Q,P}(\tau) = \text{corr}(Q(t + \tau), P(t)), \quad (19)$$

where τ —shift step of the function.

After that, the time shift at which the maximum value of the correlation coefficient is observed is determined:

$$\tau^* = \arg \max_{\tau} |r_{Q,P}(\tau)|, r_{Q,P}^* = \max_{\tau} |r_{Q,P}(\tau)|. \quad (20)$$

A high value of the maximum lagged correlation $|r_{Q,P}^*|$ at a non-zero optimal lag $\tau^* \neq 0$ indicates a delayed response of the technological stage. In this case, the absence of the maximum correlation at zero lag $\tau^* = 0$ may be explained by data imperfections, time shifts, or process inertia rather than by the ability to transfer load. If the maximum lagged correlation $|r_{Q,P}^*|$ remains moderately low even at the optimal lag τ^* , this indicates a weak relationship between the technological stage and water supply and may suggest the presence of operational flexibility in time.

3.6.3. Tariff-Directed Energy Shifts Relative to Water Supply

The F -index is a screening indicator that identifies potential zones of load transferability but does not allow a clear distinction between controllable flexibility and delayed response to water supply. To verify whether the weak coupling between energy consumption and water supply manifests itself as an economically meaningful shift of load, an analysis of the distribution of water supply and energy consumption across tariff zones is performed. The method is based on comparing the shares of water supply and the energy consumption of a technological stage within the same time intervals of the day. When a stage strictly follows water supply, the distribution of energy across tariff zones closely matches the distribution of water supply. If operational flexibility is present, energy consumption may shift toward more favorable tariff periods, such as nighttime or off-peak hours.

The total water supply and energy consumption of the i -th technological stage that fall within a selected tariff zone are determined as follows:

$$Q_z = \sum_{t \in z} Q(t), E_z = \sum_{t \in z} E(t), \quad (21)$$

where Q_{zone} , E_{zone} —the water supply and the electrical energy of the i -th technological stage within the analyzed tariff zone z .

Next, the shares of water supply and stage energy consumption within the given tariff zone are calculated relative to their total values over the entire observation period:

$$S_Q^z = \frac{Q_z}{\sum_{t \in T} Q(t)}, S_E^z = \frac{E_z}{\sum_{t \in T} E(t)}, \quad (22)$$

where S_Q^z , S_E^z —represent the shares of total water supply and energy consumption that fall within the analyzed tariff zone z .

To assess the shift of energy consumption relative to water supply, a difference indicator is introduced:

$$\Delta_z = S_E^z - S_Q^z. \quad (23)$$

The tariff-directed shift indicator Δ_z reflects the extent to which the temporal distribution of stage energy differs from the distribution of water supply. It is expected that if a stage operates strictly in synchronization with water supply, the values S_E^z and S_Q^z will be close, and therefore $\Delta_z \approx 0$.

4. Results and Discussion

4.1. Impact of Load Management Modes on the Relationship Between Water Supply and Power Consumption

The comparative analysis is based on observations of a Belarusian water source in 2019 and 2023 and a Chinese water intake facility in 2025. As part of the analysis, the Pearson correlation coefficient was calculated. This metric was used as an indicator quantifying the strength of the linear relationship between water supply and power consumption. This first step allows us to assess how closely energy consumption follows water supply at different stages of the process, over different periods, and in different systems, as well as to identify stages of water transportation where this relationship weakens. Table 5 shows the obtained correlation values by years and stages for fixing objects.

Table 5. Pearson correlation coefficient between water supply and power consumption.

Name of the Source	Year of Observations	First Water Lift	Second Water Lift	Reservoir Intake	Water Treatment Plant Purification	Pumping Station	Overall Results
Belarus	2019	0.04	0.77	—	—	—	0.52
Belarus	2023	0.08	0.95	—	—	—	0.85
China	2025	—	—	0.19	0.86	0.51	0.53

For the Belarusian water intake in 2019, the electricity consumption of the second lift pumps showed a strong relationship with the actual water supply to the urban network ($R = 0.77$), reflecting a high dependence on current demand. At the same time, the power consumption of the first lift showed almost no correlation with water supply ($R = 0.04$). This is explained by the storage function of the reservoir, as a result of which the operation of well pumps was oriented not toward instantaneous demand fluctuations but toward maintaining the required water level in the storage tank. As a result, the first lift operated relatively independently, and fluctuations in the actual water supply were only weakly reflected in changes in the load profile. As shown in Table 5, when aggregating the electricity

consumption of the two lifting stages, the weak relationship of the first stage reduced the overall correlation for the entire water intake ($R = 0.52$).

For the Belarusian water intake in 2023, a strong relationship between water supply and electricity consumption remains in the second stage, becoming even more pronounced ($R = 0.95$), while the correlation for the first stage remains close to the sign ($R = 0.08$). Given the small magnitude of R for the first lift, the difference between 0.04 and 0.08 cannot by itself be considered statistically reliable evidence of increased synchronization without additional analysis of possible time shifts. At the same time, the overall correlation for the water intake in 2023 increased significantly ($R = 0.85$), indicating a change in the consistency between the aggregated power profile and the hourly water supply profile. A possible explanation is a change in control regimes and a redistribution of the contributions of the lifting stages to total electricity consumption; however, the interpretation of this effect in the paper relies not only on the zero-lag correlation but also on lag analysis and indicators of tariff-directed load shifting.

To illustrate the relationships discussed, Figure 8 presents the correlation field between water supply and the total electrical power of Belarusian water intake facilities for 2019 and 2023. In addition, for comparison, the figure shows the correlation field for the operating regimes of the Chinese water source for 2025, as well as the linear regression equations and the corresponding trend lines for the systems under consideration. It is worth concluding that for the Belarusian and Chinese water supply systems, a clear scatter of points in the trend line area is noticeable, which indicates the presence of variability and the influence of additional factors on electricity consumption, in addition to water supply.

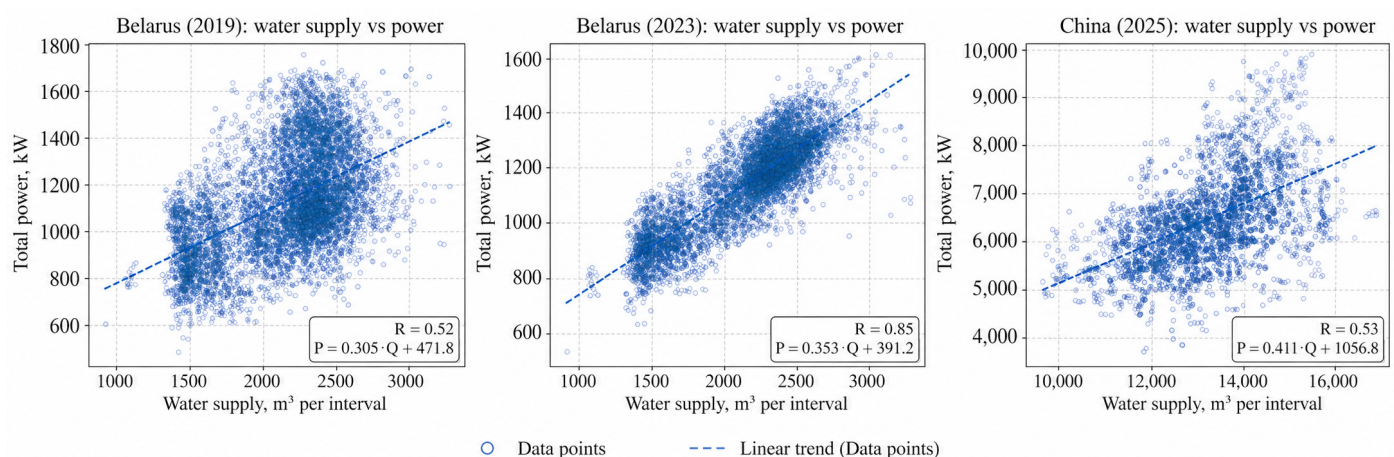


Figure 8. Correlation field and daily capacity and water consumption profiles of the Belarusian water supply system for 2019 and 2023 and the Chinese system for 2025.

Figure 9 presents the results of the lagged correlation analysis between water volume and total electricity consumption for the studied systems. The results show that the shape of the aggregated correlation curve has a pronounced daily periodicity, and the maximum correlation values for Belarus ($R = 0.85$) and China ($R = 0.53$) are observed at zero lag ($\tau = 0$). This indicates that, for both systems, the energy load is on average sufficiently synchronized with the current water supply, and no delayed effects in the operating regimes are observed.

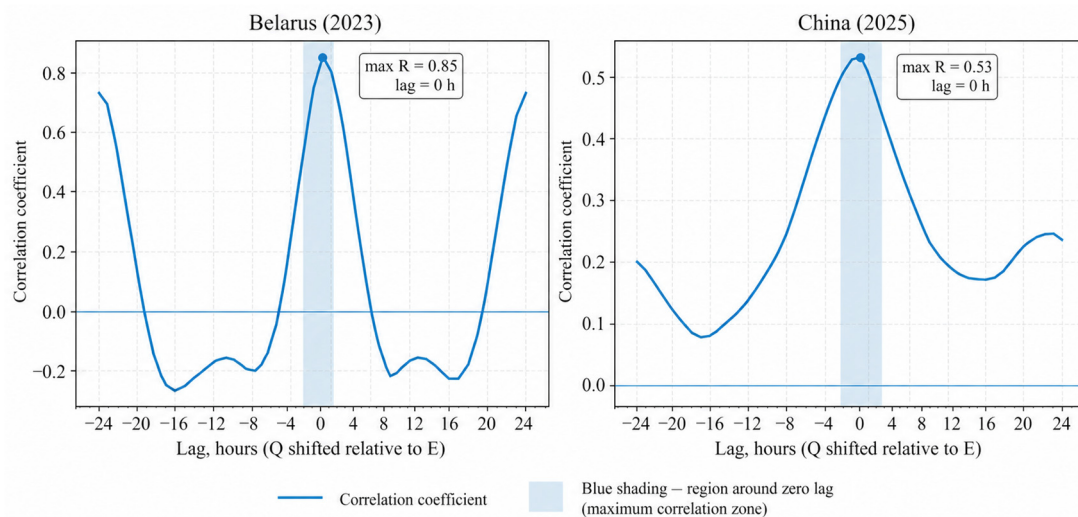


Figure 9. Lagged correlation between water supply and total electricity consumption in Belarus (2023) and China (2025).

The decomposition of lagged correlation by water supply stages makes it possible to identify which processes generate the maximum at $\tau = 0$. Figure 10 shows that the second lift of the Belarusian water source demonstrates immediate tracking of water supply: the maximum correlation is observed at zero lag with a high value of $R = 0.95$. This corresponds to an operating regime in which pump operation is determined by the current water withdrawal, and the possibilities for load shifting in time are significantly limited. In contrast, the first lift is characterized by pronounced temporal mismatch and weak coupling with water supply (the maximum correlation does not exceed $R = 0.33$ at a lag of $\tau = -6$ h).

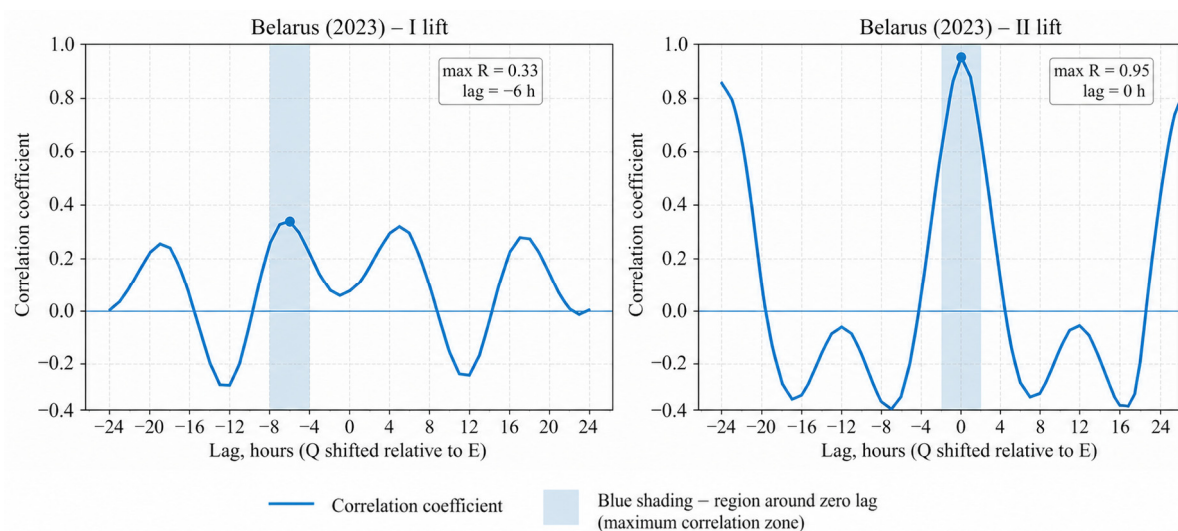


Figure 10. Lagged correlation between water supply and consumption by technological levels of electricity in Belarus (2023).

For the Chinese water supply system, the analysis shows that the maximum coupling between water supply and energy consumption at all technological stages is achieved at zero lag, meaning there is no noticeable temporal shift (Figure 11). This indicates the temporal adaptability of operating regimes, that is, when water supply changes, deviations in energy consumption occur within the same hourly interval. At the same time, the strength of the relationship between stages differs significantly: for water intake from the

reservoir the maximum correlation is $R = 0.19$; at the water treatment stage $R = 0.86$. The second-lift pumping station shows a moderate correlation ($R = 0.51$), which is consistent both with the influence of general consumer demand and with pressure regulation

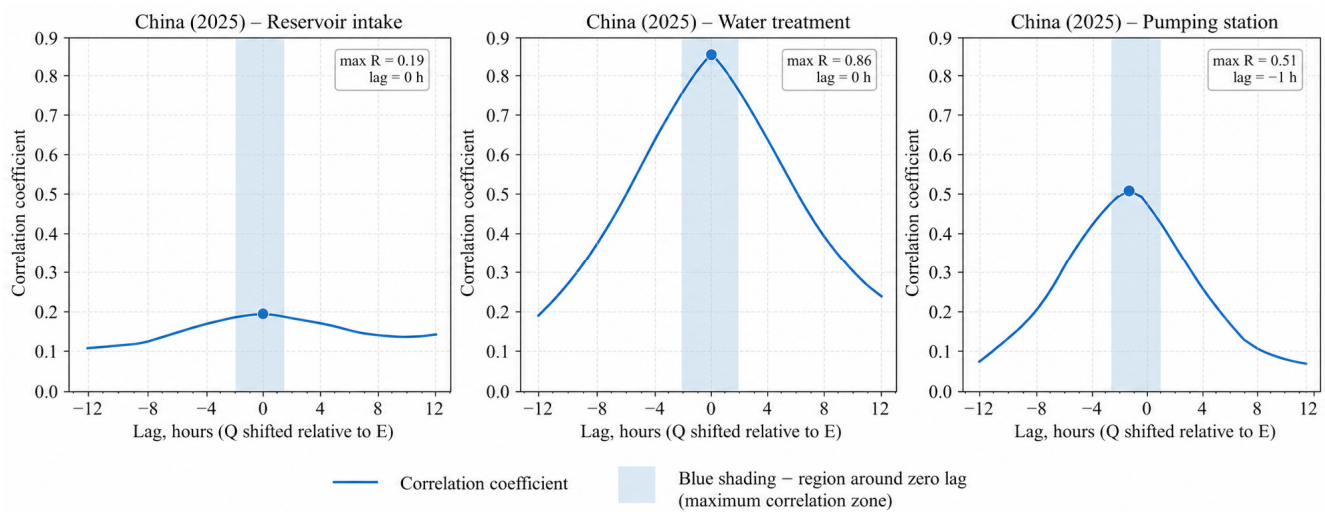


Figure 11. Lagged correlation between water supply and consumption by electricity technology levels in China (2025).

The values of the F -index (Table 6) show that the determining contribution to the integral indicator is formed by stages with a high energy share and simultaneously low correlation with water supply. For the Belarusian water source, the main source of potential load transferability is the first lift (wells): with an energy share of 69% in 2019 and 68% in 2023, the correlation with water supply remains close to zero (0.04 and 0.08, respectively), which ensures the main contribution to the F -index at the level of 0.66 and 0.63. In contrast, the second lift (water intake) forms only a small contribution to the F -index (0.07 and 0.02). As a result, the integral indicator for the Belarusian facility decreases from 0.73 in 2019 to 0.65 in 2023, indicating an increase in the consistency of the aggregated power profile with water supply and reflecting a more active use of dispatch control of pumping unit operating modes compared with 2019.

Table 6. F -index of load transfer.

Dataset	Stage	Energy Share of Stage, %	Correlation	F -Index of Stage
China (2025)	Reservoir intake	38%	0.19	0.31
China (2025)	Water treatment	24%	0.86	0.03
China (2025)	Pumping station	38%	0.49	0.19
China (2025)	Resulting F-index			0.53
Belarus (2019)	Wells	69%	0.04	0.66
Belarus (2019)	Water intake	31%	0.77	0.07
Belarus (2019)	Resulting F-index			0.73
Belarus (2023)	Wells	68%	0.08	0.63
Belarus (2023)	Water intake	32%	0.95	0.02
Belarus (2023)	Resulting F-index			0.65

For the Chinese system, the highest contribution to the F -index is formed by the water intake stage from the reservoir: with an energy share of 38% and a low correlation of 0.19, the F -index component equals 0.31, indicating the presence of potential for operational control. The pumping station (second lift), with a comparable energy share of 38%, demonstrates a moderate correlation of 0.51 and forms a contribution of 0.19. The water treatment

stage has a high correlation of 0.86 with an energy share of 24% and therefore contributes almost nothing to the potential (F -component = 0.03). The resulting value of the F -index for China is 0.53, which is lower than the corresponding value for Belarus.

Overall, the comparative analysis shows that for the Belarusian system the potential load transferability is most pronounced at the first lift stage. For the Chinese system, the largest reserve of weak coupling is observed at the reservoir water intake stage, whereas water treatment remains the most synchronized stage. At the same time, the interpretation of moderate F -index values for the second-lift pumping station should take into account the influence of pressure and hydraulic conditions, which can reduce linear correlation without the presence of controllable time shifting. Therefore, further justification of transferability for pumping stages should be supported by tariff-directed indicators of load shifting.

4.2. Comparative Analysis of Daily Water and Electricity Consumption Profiles

The comparison of operating regimes of the Belarusian and Chinese water supply systems is based on normalized typical daily profiles of water supply and energy consumption and is presented in Figure 12. The obtained daily water consumption profile for the Belarusian water intake is characterized by pronounced irregularity and a decrease in intake loading during the night hours (from 02:00 to 05:00), followed by a sharp increase in consumption in the morning period from 06:00 to 11:00 and a clearly formed evening maximum from 18:00 to 22:00. The pronounced bimodality of consumption leads to variations in operation within a wide range, from 0.69 to 1.19 of the average value. The distribution of the hours of occurrence of the daily maximum water demand also confirms these observations: on average, 46.3% of daily peaks occur at 21:00, i.e., in the late evening. For the Chinese water supply system, the water consumption profile is more smoothed and varies only slightly relative to the mean, within 0.97–1.02. At the same time, the average maximum does not show a strong concentration and is distributed both in the daytime hours (13:00 accounts for more than 11.1%) and in the evening period (from 16:00 to 23:00).

Active power consumption generally follows the shape of water consumption in both systems; however, for the Belarusian water intake some differences are observed in variability relative to the average value, within 0.80–1.12, which is explained by the presence of conditionally constant electricity consumption that does not depend on the volume of water supply. The average load maximum is also concentrated in the evening hours (22:00), but with a greater shift relative to the peaks of water supply and a lower concentration of occurrence frequency (25.5%). For the Chinese system, a smaller amplitude of power fluctuations is characteristic, while the concentration of maximum values is observed mainly in the evening hours (21:00–22.2%).

The observed differences in Figure 12 reflect the characteristics of the daily demand patterns in the studied systems. For the Belarusian facility, the pronounced late-evening maximum is associated with the influence of industrial load and the daily activity of the population, whereas for the Chinese facility, the smoother profile and the more evenly distributed maxima indicate operation under a more stable water supply regime. The shift of the active power maximum relative to the water supply maximum in the Belarusian system (22:00 versus 21:00) can be considered an indirect indicator that electricity consumption is determined not only by the current demand but also by operational modes and load management elements, which results in a partial mismatch between the peaks of water supply and energy consumption. For the Chinese system, the concentration of water demand maxima is less pronounced and occurs more frequently during daytime hours, while the maxima of active power are shifted toward the evening–night period.

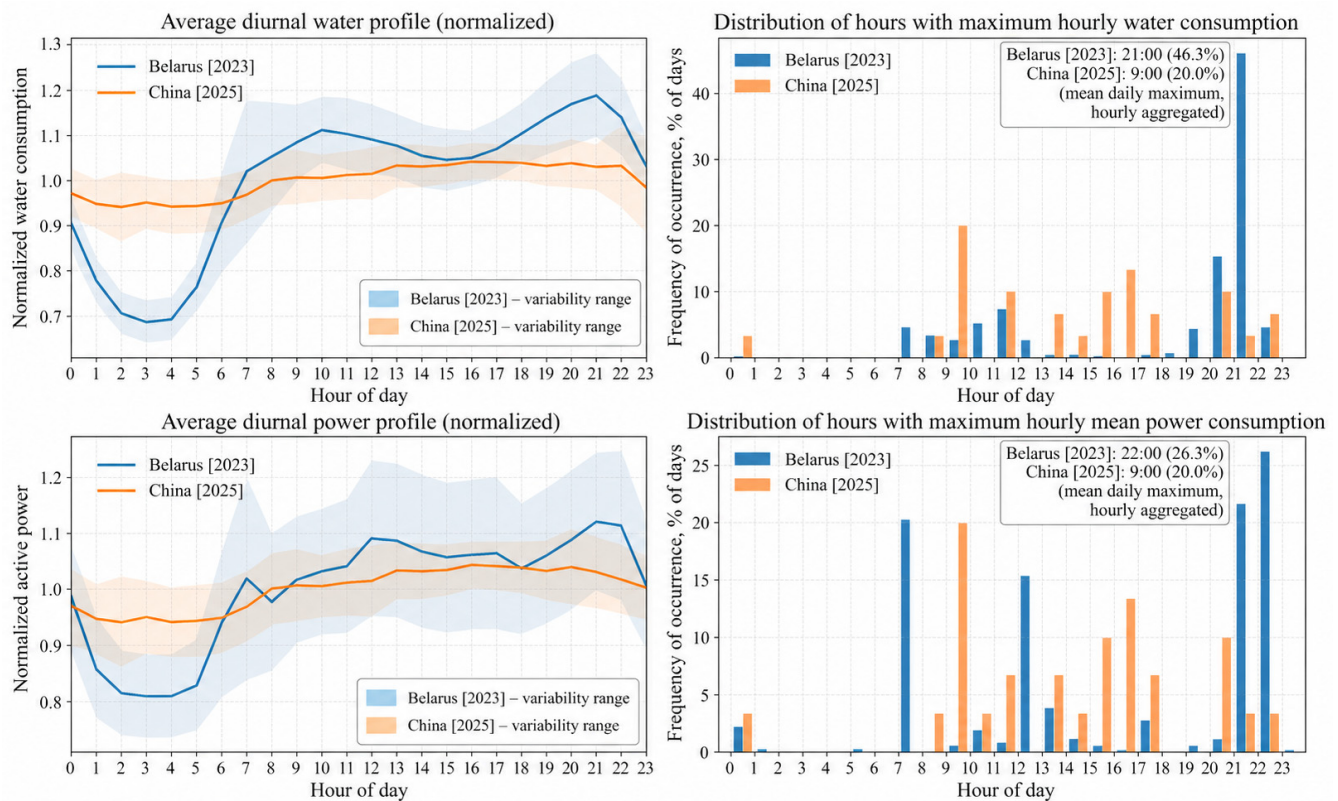


Figure 12. Comparison of water supply and electricity consumption profiles with the distribution of average power peaks by daytime zones of the Belarusian and Chinese water supply systems.

4.3. Assessment of Tariff-Driven Energy Shifting

From the perspective of capacity payment and assessing the feasibility of transitioning to a time-zone-based energy payment system, an analysis of the frequency distribution of absolute daily peaks is of particular interest. Figure 13 shows a daily diagram of absolute load peaks and the probability of their occurrence during the day. For the Belarusian water supply system, daily peaks are overwhelmingly concentrated in off-peak zones, demonstrating organized efforts (in 2023) to reduce load during peak hours of the power grid. In contrast, for the Chinese system, a significant share of peaks is concentrated during peak intervals (61.8%), compared to 3.3% for the Belarusian source. This distribution reflects the significant role of tariff policy: in Belarus, capacity charges are determined by the highest load value recorded during peak hours of the power grid, whereas in China, the calculated capacity is determined by the absolute maximum load during the day, regardless of the time of its occurrence.

Additionally, to the analysis of daily peak occurrences, Table 7 presents the results of tariff-oriented shifting of electricity consumption relative to water delivery, quantified by the Δ_z indicator. For the Belarusian system, the results are consistent with the displacement of maxima outside the power-system peak window: at the first lift (wells), a stable shift of the load toward the night zone is observed, with $\Delta_{night} = 5.69\%$ in 2019 and 6.02% in 2023, along with a slight reduction in the load in the peak zone, with $\Delta_{peak} = -1.06\%$ to -1.70% . This indicates that the electricity consumption of the first lift is shifted toward more advantageous time intervals to a greater extent than water delivery, which can be interpreted as a result of targeted operational control. In the Chinese system, fundamentally different patterns are observed: for reservoir intake and water treatment, the Δ_z values in the night zone are close to zero, indicating weakly expressed tariff-driven load shifting.

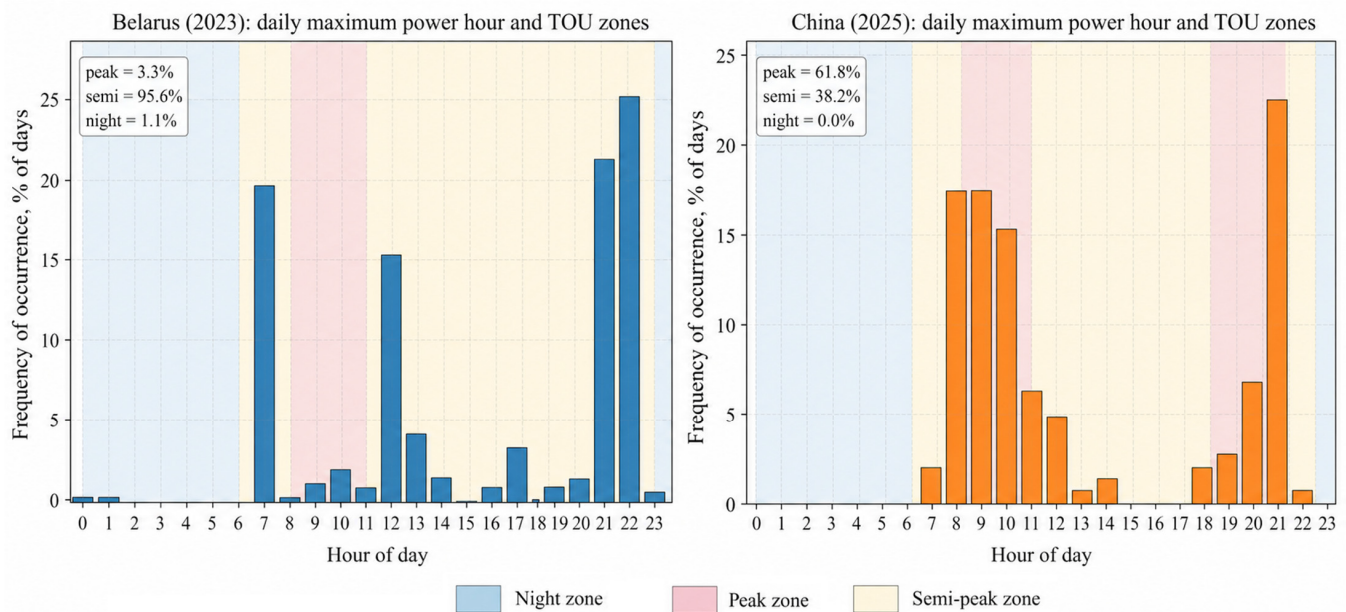


Figure 13. Frequency distribution of absolute daily maximums across TOU (Time-of-Use) tariff zones.

Table 7. Tariff-oriented load shifting by technological stages.

Object	Technological Stage	Δ Night, %	Δ Peak, %
Belarus (2019)	Wells (I lift)	5.69	−1.06
Belarus (2019)	Water intake (II lift)	−6.91	1.83
Belarus (2023)	Wells (I lift)	6.02	−1.70
Belarus (2023)	Water intake (II lift)	−6.02	0.88
China (2025)	Reservoir intake	0.33	−0.10
China (2025)	Water treatment	−0.26	0.19
China (2025)	Pumping station	−7.02	3.78

For the second-lift pumps, the same pattern is observed in both systems: in Belarus (2023), the values amount to -6.02% in the night zone and $+0.88\%$ during peak hours, whereas in China (2025) the corresponding values are -7.02% and $+3.78\%$. This indicates that during periods of maximum water demand the second lift becomes more energy-intensive due to the increase in discharge pressure; as a result, the load is concentrated in peak tariff intervals.

4.4. Impact of the Tariff Model on the Economic Efficiency of Water Intake Electricity Consumption Management

This section presents the results of the electricity load redistribution experiment at a Belarusian water source for 2019 and 2023, achieved through the implementation of the adopted electricity consumption management strategy under the current two-part tariff model. The analysis was conducted to assess changes in the shape of the daily power profile and the shift of half-hourly peaks from high-priority hours. Figure 14 shows changes in the frequency of peaks across tariff intervals.

As a result of the implemented measures, a targeted redistribution of half-hourly power peaks across tariff zones is observed. Figure 14 shows that in 2023, compared to 2019, the frequency of peaks in the morning peak zone (08:00–11:00) and the evening peak interval (18:00–21:00) has significantly decreased, while their concentration in the semi-peak zone (06:00–08:00 and 11:00–23:00) has increased. The identified changes are consistent with the profile of the daily average power and allow us to conclude that the key control regulator is the pumping equipment of the first water lift. Moreover, for the most concentrated time in

terms of load, 22:00, the average contribution of the first and second water lift was 778 kW to 503 kW. The capacity ratio of 1.55 confirms that the position of the daily peak is determined by the well pump schedule. Figure 15 shows the shape of the first lift load graph after switching off some of the pumps during peak hours of the power system.

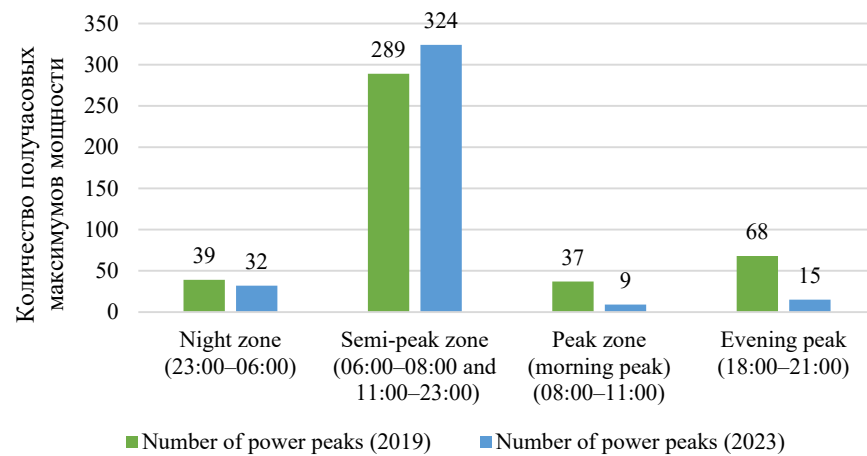


Figure 14. Distribution of annual frequency of occurrence of half-hourly maxima before and after load regulation.

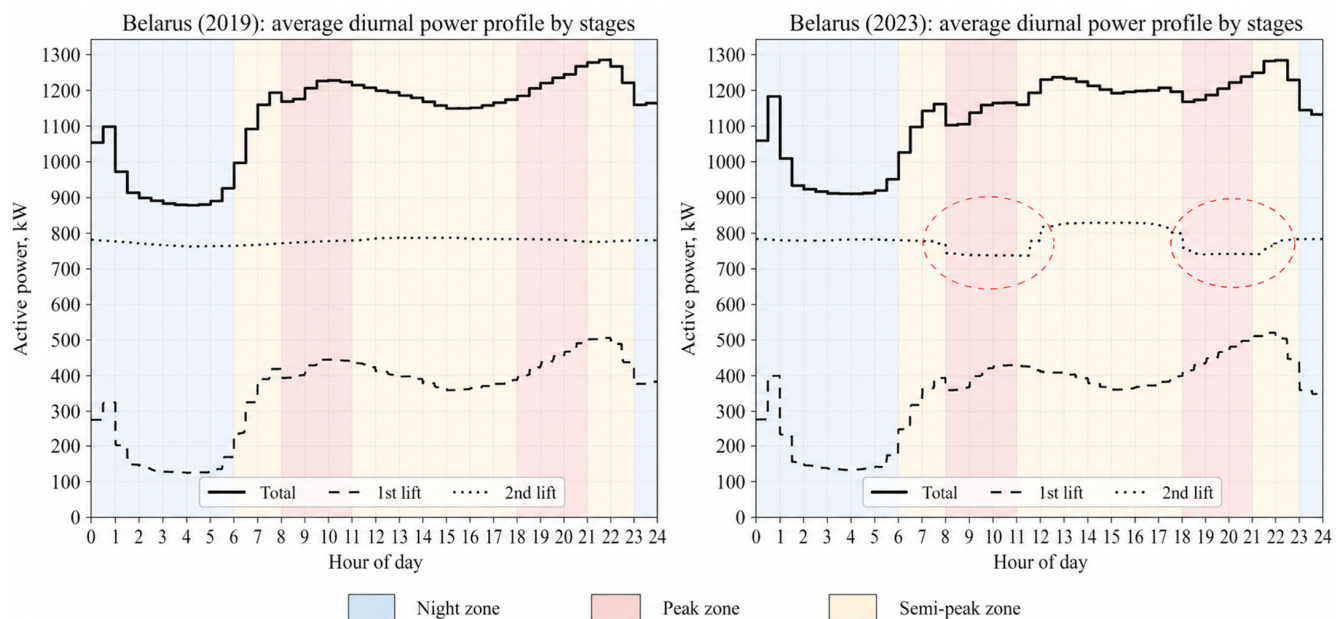


Figure 15. Average daily profile of half-hourly active power of the Belarusian water source before (2019) and after (2023) adjusting the schedule for switching on the first water lift pumps.

The second peak follows water consumption, with peak water consumption occurring in the evening from 6:00 PM to 9:00 PM (more than 40%), while the half-hourly maximum is concentrated between 9:30 PM and 10:00 PM, i.e., formally in the semi-peak tariff zone. This demonstrates that the first peak's electricity demand management reduced the impact of the absolute maximum evening water demand. In 2023, when the maximum water consumption occurred during the evening peak of power system loads, in 88% of such cases, the highest total capacity was recorded in the semi-peak zone. The average daily profiles (Figure 15) show a reduced impact of the evening peak in water consumption on the total water intake capacity. In 2019, the profile shape had a pronounced morning and evening peak, coinciding with the increase in demand. In 2023, due to an increase in the first-lift load for filling reservoirs during daytime hours ($\approx 12:00\text{--}16:30$) and controlled activation

during the evening water demand period (22:00), the overall power profile became more uniform in the morning (08:00–11:00) and evening (18:00–21:00) periods, but increased in the semi-peak zone (the share of the three most frequent half-hourly peaks increased from 56% to 61%, and the share of the five peaks from 68% to 77%). This indicates that the moment of average half-hourly power has become more predictable, and the scheduling of modes allows it to be kept within a controllable zone.

The result of the implemented organizational measures demonstrates a clear economic potential. With equal annual electricity consumption (the 2019 data were normalized to the 2023 level), the total electricity payment decreased by 1.58%, which corresponds to approximately USD 23.3 thousand (the calculations were performed in Belarusian rubles (BYN) with subsequent conversion using an average exchange rate of ≈ 3.2 BYN/USD). The calculation was carried out using the applicable two-part tariff, which includes payment for maximum power demand and consumed electricity. The following tariff parameters were used: the capacity tariff $a = 30.08196$ BYN/kW per month, the energy tariff $b = 0.25439$ BYN/kWh, and the calculations also included VAT (20%) and a currency adjustment coefficient of 1.076. The main source of the economic effect was the reduction in the capacity charge, which decreased by 8.6%, while the energy charge remained unchanged. As a result, the share of the capacity component in the total electricity payment decreased from 18.2% to 16.9%. This is a direct consequence of shifting the half-hour peak demand outside the morning and evening system peak windows. It should be noted that the achieved cost reduction represents the maximum attainable effect under the applied operational approach, since the potential of optimization-based planning of pump operation, considering hydraulic constraints and price signals, was not utilized. The implemented measures were limited to organizational actions aimed at reducing power demand during system peak hours, which led to the reduction in the capacity payment.

To compare the obtained results, Figure 16 shows the average daily active power profile of the Chinese water source, broken down by process stages. Unlike the Belarusian system, the daily power profiles are characterized by the absence of distinct intra-day modes at individual process levels. Thus, for the Chinese tariff model, the economic efficiency of electricity consumption management is largely achieved through a general limitation of the absolute maximum load.

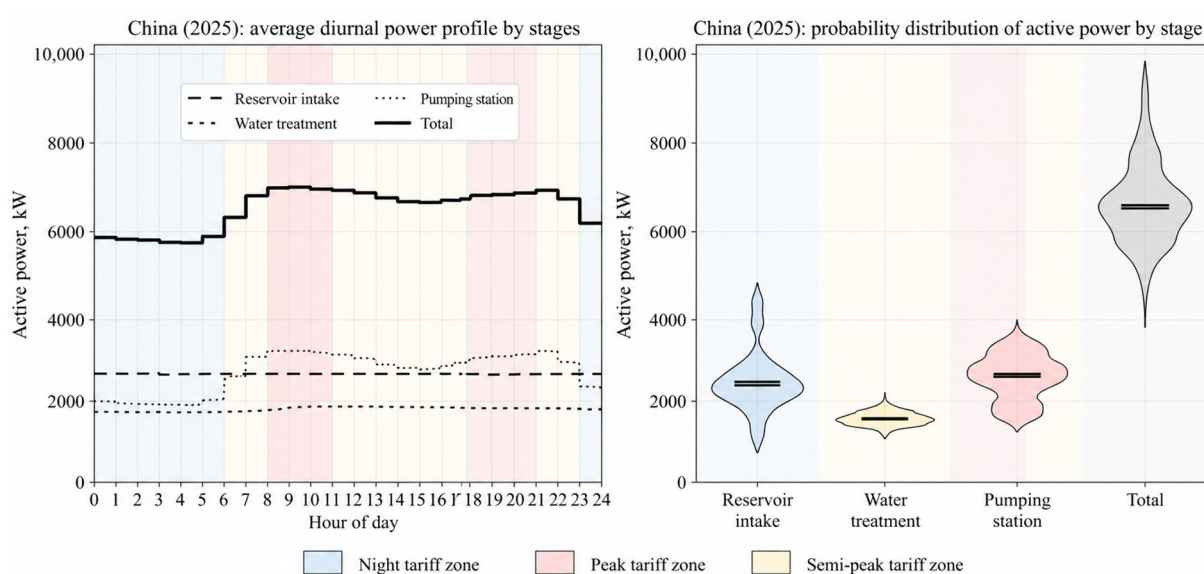


Figure 16. Average daily profile of hourly active power of a Chinese water source by levels of technological stages.

4.5. Prerequisites for Planning Pump Activation Modes in Conditions of Tariff Differentiation

For the Belarusian water source, the existing two-part tariff model has proven effective in reducing peak loads by disconnecting some consumers during peak hours. The anticipated strategy for developing the electricity consumption management system is a transition to an alternative billing system based on time zones. To determine the prerequisites for such a transition, the most significant time intervals for peak load generation in Belarus were identified: morning (8:00–11:00) and evening (18:00–21:00) peaks. The calculations used half-hourly average power values, with the specified time corresponding to the end of the calculation interval (e.g., the value 8:30 reflects the average load for the period 8:00–8:30). The same logic was applied to determining the tariff zones established by the electricity differentiation instructions: night (11:00 PM–6:00 AM), peak (8:00 AM–11:00 AM), and semi-peak (the rest of the day). Figure 17 shows the changes in daily maximums of average half-hourly active power by hour of the day for 2019 and 2023.

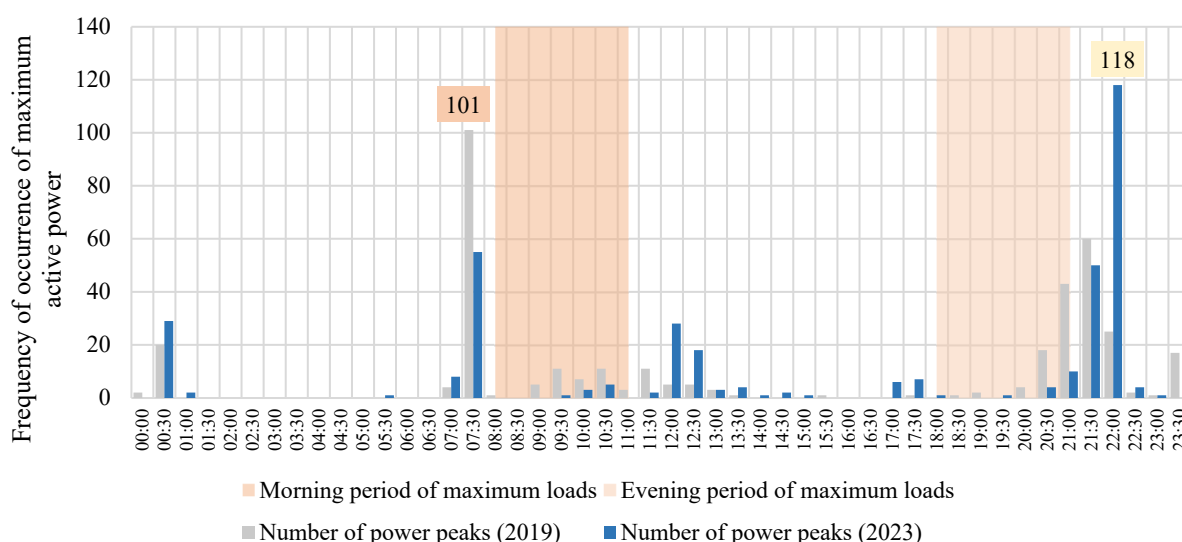


Figure 17. Distribution of daily absolute maximums of half-hourly active capacity of the Sozh water intake by half-hour for 2019 and 2023.

It can be noted that in 2019, the daily maximums of the total capacity of the Belarusian water intake had two distinct peaks: at 07:30 (morning) and 21:30 (evening). In 2023, after the implementation of the borehole pump activation management strategy, a shift and concentration of the maximum at 22:00 was observed (118 days out of 365), meaning that it was at this time that the highest half-hourly active capacity was most often recorded. Across tariff zones, this was accompanied by a displacement of peaks from the morning and evening periods to the semi-peak zone: the share of days with a maximum coinciding with the system peak range of 08:00–11:00 decreased from ~9.9% to ~2.5%, and in the interval of 18:00–21:00—from ~6.8% to ~1.6%. At the same time, the frequency of peaks in the semi-peak zone increased from ~72.6% to ~86.8%. Thus, the company has virtually ceased generating absolute daily peaks in the intervals taken into account when calculating payments under the current two-part (non-differentiated by daytime zone) tariff.

The Belarusian electricity pricing model allows for payment at a two-part differentiated tariff only if a key condition is met: enterprises are required to schedule their schedules so that the evening peak does not exceed the morning peak; otherwise, payment is made at the two-part tariff. In 2019 (Figure 18a), more than 61% of days had evening peak power exceeding the morning peak, making it difficult to meet this condition. In 2023 (Figure 18b), the proportion of such days increased to 82.5%, but the distribution of values

became significantly more clustered, and the dispersion decreased. This suggests that the regulation strategy has shifted its emphasis from variability to predictability.

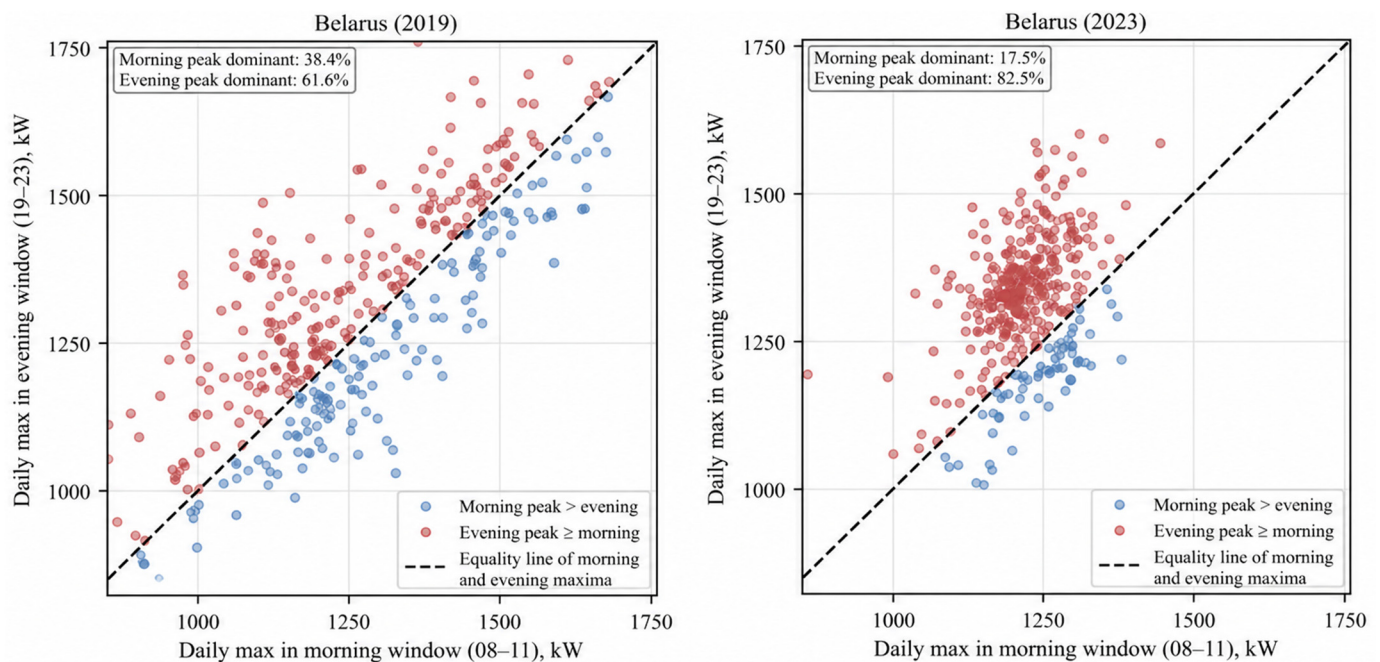


Figure 18. Comparison of morning and evening power peaks for 2019 (a) and 2023 (b).

For days where $\Delta_d > 0$ (the evening peak is higher than the morning peak), a distribution was constructed with the load ranked in ascending order. As a result, the data analysis revealed that in 2019, the evening peak exceeded the morning peak by an average of 96.7 kW, while in 2023, it exceeded the morning peak by 85.6 kW. Therefore, to meet the differentiation condition, it is sufficient to minimally reduce the evening local maximum by 119 kW (on 80% of days), 156 kW (on 90% of days), and 191 kW (on 95% of days) among those cases where the evening peak initially exceeded the morning peak. If the morning peak is maintained, meeting the required condition will require temporarily shutting down approximately 7–8 borehole pumps during problematic half-hour intervals. Two existing reservoirs with a total capacity of 30,000 m³ can be used to compensate for the reduced water production volumes. These results show that the adopted electricity consumption management strategy generally reduces charges under the two-part tariff but does not ensure a minimum. To enhance the economic impact of switching to a differentiated payment system, more precise planning algorithms are needed that take into account technological variability and future water demand, which is part of further research.

5. Conclusions

The results of the study showed that the effectiveness of electricity-demand management at a water intake is determined not only by the technological configuration itself, but also by how strongly the process scheme and the water-withdrawal regime allow equipment operation to be shifted within the day. A comparison of the Belarusian and Chinese water-supply systems revealed fundamental differences in the mechanisms governing daily peak demand and the redistribution of load across tariff time bands. A two-part tariff with predefined power-system peak hours creates conditions for actively shaping the daily load profile. The implemented energy consumption management strategy for the Belarusian water intake resulted in a shift of half-hourly power peaks from morning and evening peaks to semi-peak zones, resulting in an 8.6% reduction in capacity charges. This outcome should be interpreted as an effect of organizational and dispatch control –namely,

scheduling well operation—within existing technological constraints. At the same time, managing electricity consumption without explicitly accounting for tariff zones increased the share of cases in which the evening maximum exceeds the morning maximum, which is critical under time-differentiated electricity pricing. Therefore, when transitioning to time-of-use pricing, a key practical requirement is the explicit incorporation of tariff intervals and peak-load control into the operational control rules.

The assessment of controllability in the Chinese water-supply system indicates a higher degree of adaptation of operating modes to the current water demand. Under these conditions, the primary focus is not so much on shifting the load within the day, but rather on improving energy efficiency at the pumping-station level and on managing storage in reservoirs within admissible operating limits. The diagnostic metrics used in this study—lagged correlation, the F-index, and tariff-oriented shifting indicators—make it possible to identify promising stages for electricity-demand management. At the same time, this work does not formulate an optimization problem, a control-system architecture, or a complete set of operational constraints; developing and validating these elements on practical data are considered promising directions for further research. In addition, a further research avenue is to investigate how urban water-supply systems can adapt their energy-management strategies to electricity price signals in real time.

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